K J SOMAIYA INSTITUTE OF TECHNOLOGY
Department of Information Technology

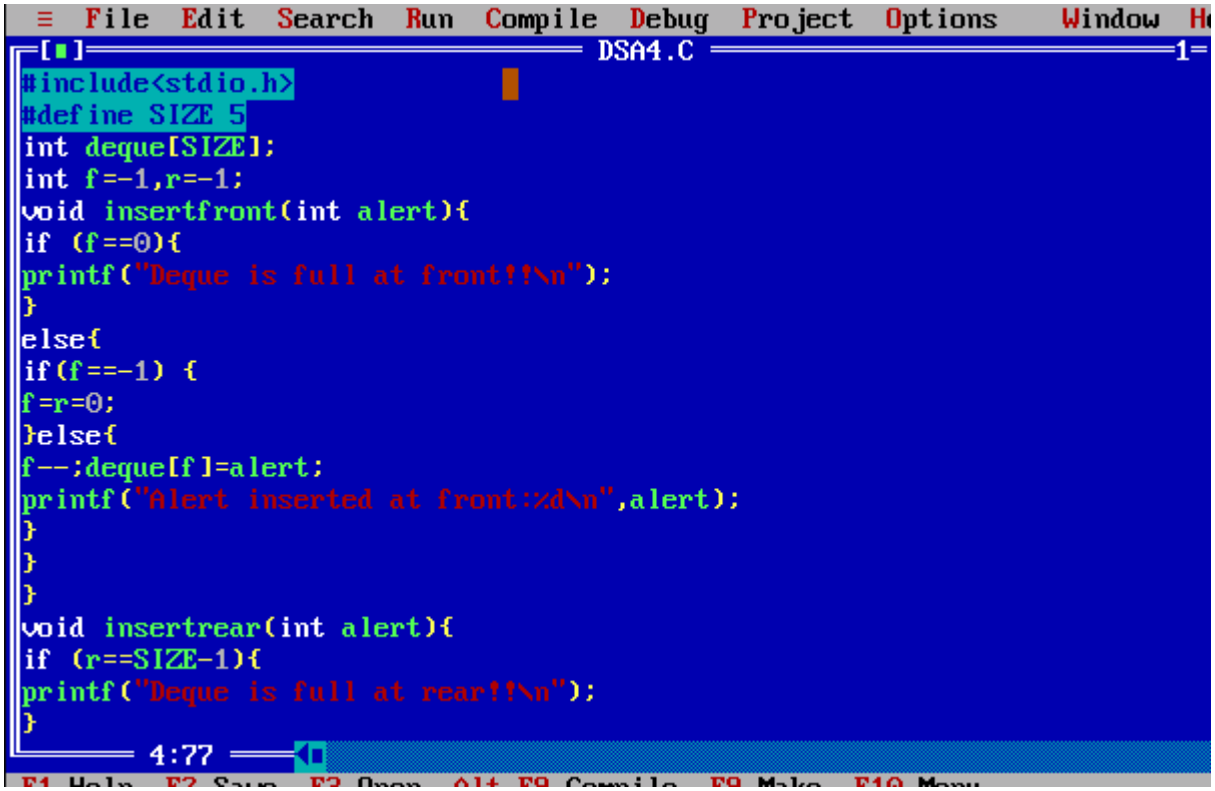
Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

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Batch: SA2
Roll Number: 27

EXP NO:04

Program:




```
void insertrear(int alert){
if (r==SIZE-1){
printf("Deque is full at rear!!\n");
}
else{
if(f==-1){
f=r=0;
}else{
r++;
deque[r]=alert;
printf("Alert inserted at rear:%d\n",alert);
}
}
}
```

```
void deletefront(){
if(f==-1){
printf("No active heatwave alerts!!\n");
}
else{
printf("Alert removed from front!!:%d\n",deque[f]);
if(f==r)
f=r=-1;
else
f++;
}
}
```

```
void deleterear(){
if(r==-1){
printf("No active heatwave alerts!!\n");
}
else{
printf("Alert removed from rear!!:%d\n",deque[r]);
if(f==r)
f=r=-1;
else
r--;
}
}
```

```
void display(){
int i;
if(f==-1){
printf("No active heatwave alerts!!\n");
}
else{
printf("Active Heatwave Alerts:");
for(i=f;i<=r;i++)
printf("%d\n",deque[i]);
printf("\n");
}
}
```

```c
int main(){
int choice,temp;
while(1){
printf("\n----Heatwave Monitoring System----\n");
printf("1.Add alert at front\n");
printf("2.Add alert at rear\n");
printf("3.Remove alert from front\n");
printf("4.Remove alert from rear\n");
printf("5.Display\n");
printf("6.Exit");

printf("\nEnter Choice:");
scanf("%d",&choice);
switch(choice){
case 1:
printf("Enter Temperature:");
scanf("%d",&temp);
insertfront(temp);
break;
```

```c
case 2:
printf("Enter Temperature:");
scanf("%d",&temp);
insertrear(temp);
break;
case 3:
deletefront();
break;
case 4:
deleterear();
break;
case 5:
display();break;
case 6:
return 0;
default:
printf("Invalid choice!!\n");
}
}
}
```

Output:

```
----Heatwave Monitoring System----
1.Add alert at front
2.Add alert at rear
3.Remove alert from front
4.Remove alert from rear
5.Display
6.Exit
Enter Choice:1
Enter Temperature:34
```

```
----Heatwave Monitoring System----
1.Add alert at front
2.Add alert at rear
3.Remove alert from front
4.Remove alert from rear
5.Display
6.Exit
Enter Choice:2
Enter Temperature:20
Alert inserted at rear:20

----Heatwave Monitoring System----
1.Add alert at front
2.Add alert at rear
3.Remove alert from front
4.Remove alert from rear
5.Display
6.Exit
Enter Choice:3
Alert removed from front!!:0

----Heatwave Monitoring System----
1.Add alert at front
2.Add alert at rear
3.Remove alert from front
4.Remove alert from rear
5.Display
6.Exit
Enter Choice:5
Active Heatwave Alerts:20


----Heatwave Monitoring System----
1.Add alert at front
2.Add alert at rear
3.Remove alert from front
4.Remove alert from rear
5.Display
6.Exit
Enter Choice:5
Active Heatwave Alerts:20
```